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PUBLISHED ONLINE: 21 OCTOBER 2012 | DOI: 10.1038/NMAT3451

nature materials

Long-range interlayer alignment of intralayer domains in stacked lipid bilayers

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Liquid-crystalline phases of stacked lipid bilayers represent a pervasive motif in biomolecular assemblies. Here we report that, in addition to the usual smectic order, multicomponent multilayer membranes can exhibit columnar order arising from the coupling of two-dimensional intralayer phase separation and interlayer smectic ordering. This coupling propagates across hundreds of membrane lamellae, producing long-range alignment of phase-separated domains. Quantitative analysis of real-time dynamical experiments reveals that there is an interplay between intralayer domain growth and interlayer coupling, suggesting the existence of cooperative multilayer epitaxy. We postulate that such long-range epitaxy is solvent-assisted, and that it originates from the surface tension associated with differences in the network of hydrogen-bonded water molecules at the hydrated interfaces between the domains and the surrounding phase. Our findings might inspire the development of self-assembly-based strategies for the long-range alignment of functional lipid domains.

Multilamellar lyotropic phases of stacked lipid bilayers exhibiting smectic liquid-crystalline order[1] (or membrane multilamellae) are spontaneously produced primarily through hydrophobic forces resulting from lipid–water interactions. Continued interest in these membrane multilamellae can be attributed to their relevance in many areas of fundamental research and practical applications. In living systems, such multilamellar membranous structures are present as lamellar bodies[2]. They also occur in many active biological assemblies that control energy transduction. For instance, thylakoid membranes of photosynthetic cyanobacteria[3] or plant chloroplasts[4], and electrocyte cells in electric eels[5] are highly folded membranous structures that combine single membrane functions in series, which because of their energy-harvesting capabilities hold considerable technological potential for biomimicry[6]. Moreover, potential for applications in photonics, sensing and DNA delivery (for example, gene therapy and synthetic biology), further fuels interest in these liquid-crystalline multilamellar membranes[7–9].

A unique feature of such smectic mesophases of stacked-membrane bilayers is that although the constituent lipids exhibit in-plane translational degrees of freedom, their out-of-plane incoherent undulation fluctuations are severely restricted because of the steric spatial confinement they experience with the neighbouring layers[1]. It is now well appreciated that the entropic loss associated with these so-called Helfrich intermembrane repulsive interactions stabilizes the one-dimensional (1D) smectic order of uniform lipidic mesophases (that is, 1D positional order).

In the past, studies of multilamellar membranes have focused dominantly on uniform systems consisting of single lipids. Biological membrane mesophases and those relevant to practical biomimetic applications, however, are seldom uniform. Multicomponent mixtures of lipids and proteins, which readily phase-separate into coexisting phases, characterize these functional membrane multilamellae. This raises the possibility that two-dimensional intralayer phase separation of lipids and proteins in individual membrane lamellae couples with the interlayer smectic ordering to align functional components across membrane lamellae in a self-organized manner. However, such higher-order organization in multilayered membranes has not yet been demonstrated.

Here, we show that the phase-separated domains in individual layers of the multilamellar membrane stacks align with one another, spontaneously producing three-dimensionally aligned domains across hundreds of lamellae. Many classes of functional domains (for example, cholesterol (Chol)- and sphingomyelin (SM)-enriched liquid-ordered, L_o, domains or lipid rafts) in biological membranes are responsible for sensing and signalling transport activities[10–13]. Such domains selectively partition many membrane proteins and thus serve as functional hotspots in single bilayers. For instance, a variety of ion channels such as K^+ (ref. 14), Na^+ (ref. 15) and Ca^+ channels[16] are known to localize in L_o domains. Therefore, the alignment (registration) of domains should also align the ion channels. Our observations that these hotspots can spontaneously align or register across hundreds of single bilayers suggest possibilities for serial coupling of their functions (for example, transport) for practical applications.

The membrane multilamellae we investigated consist of putative raft-forming ternary lipid mixtures composed of Chol, SM, and an unsaturated phospholipid, DOPC (1,2-dioleoyl-sn-glycero-3-phosphocholine). Membrane multilamellae on solid silica substrates are prepared by adapting previously reported techniques[17,18]. Sample thickness was varied between 1 and 7 μm (measured by depth-resolved confocal fluorescence imaging),

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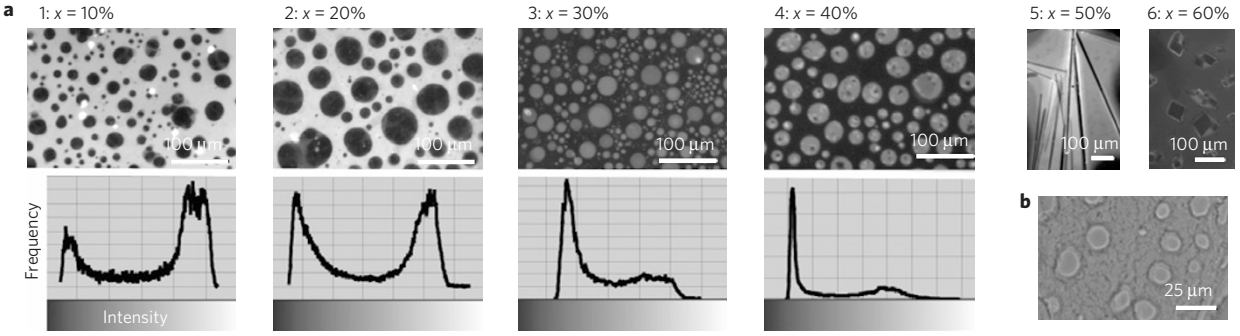

Figure 1 | Composition-dependent phase separation in stacked lipid bilayers. a, Epifluorescence images for stacked lipid bilayers consisting of equimolar proportions of DOPC and SM and a variable concentration (designated as x) of Chol (1:1 SM/DOPC + x = 10–60% Chol) in bulk water. Samples are doped with 0.1% Rho-PE. For all Chol concentrations up to 40%, coexisting L_o and L_d phases are observed (1–4). The intensity histograms of the images provide a comparison of the relative areal populations of the black (L_o) and white (L_d) regions, revealing a monotonic trend consistent with Chol partitioning into the L_o phase. For x = 50% and 60% crystals of Chol are evident (5,6). **b**, A bright-field image of domains in a sample (x = 10%) prepared without the fluorescent probe lipid.

representing approximately 150–1,200 bilayers, simply by varying lipid concentrations and deposition conditions. For study, we systematically varied Chol concentration in equimolar SM and DOPC mixtures (1:1 SM/DOPC + xChol; x = 10–60% (molar)). When organized as single bilayers (for example, giant unilamellar vesicles) below their miscibility transition temperature, these lipid mixtures are known to phase-separate into two coexisting liquid phases, one enriched in Chol and SM designated as L_o (liquid-ordered) phase, and a second L_d (liquid-disordered) phase consisting primarily of unsaturated DOPC (refs 19–22). To discriminate between the L_o and L_d phases by fluorescence microscopy, the lipid mixtures were doped with phase-sensitive naphthopyrene (Naph) and/or N-lissamine rhodamine palmitoyl phosphatidylethanolamine (Rho-PE), which partition preferentially into the L_o and L_d phases, respectively[23,24]. We performed experiments in bulk water as well as under a range of partial relative humidity spanning ~30–100%, and monitored 3D domain alignment using wide-field and confocal fluorescence microscopy techniques. Although the dynamics of domain alignment was slower at lower humidities, phase-separated morphologies comparable to those obtained in bulk water ultimately formed in all cases. Unlike multilamellae immersed in bulk water, for which long-term structural integrity is limited (<1 h), those maintained under partial humidity conditions exhibited long-term stability (tested for over a year).

Phase separation in stacked lipid bilayers

Epifluorescence images establish that the membrane multilamellae formed from these mixtures produce coexisting L_o and L_d phases for Chol concentrations up to 40 mol% [1:1 SM/DOPC + xChol; x = 10–40%]. Representative examples are shown in Fig. 1a(1–4), in which the Rho-PE probe provides a distinction between the two phases. For all samples, nearly circular domains (occasionally elongated), several tens to hundreds of micrometres in diameter, are seen to decorate the images. For low-Chol-content samples (for example, x = 10%), the domains appear essentially black (B) for the L_o phase surrounded by the brighter (W) background for the L_d phase. As expected, this binary B/W fluorescence pattern becomes inverted (W/B) for samples with higher Chol content (x = 40%). For x = 20–30%, both B/W and W/B configurations, as well as nested domains (smaller domains embedded within larger domains of the opposite kind), are observed. Corresponding intensity histograms in Fig. 1 confirm these phase assignments by revealing a monotonic rise in B (L_o phase) regions for higher Chol concentrations. Replacing Rho-PE with Naph yields complementary fluorescence patterns (data not shown), lending further support to the L_o and L_d phase assignments. Bright-field optical images reveal comparable domains in samples devoid of any fluorescent probes, confirming that the domains observed in fluorescence microscopy are not influenced by the presence of the dye (Fig. 1b). The shapes and sizes of these domains are qualitatively similar to those observed for free single lipid bilayers organized as giant unilamellar vesicles[25]. For higher Chol concentrations (x > 40%), this liquid–liquid coexistence is abandoned, and macroscopic crystals of Chol, consistent with those reported earlier[26], become evident (Fig. 1a(5,6)).

It is notable that, despite multilamellarity, the fluorescence intensity histograms (Fig. 1) reveal binary spatial patterns, which provide an initial clue for the possibility that the domains align across individual lamellae. To better examine interlamellar domain alignment, we investigated the depth profile (z-slices) of samples using confocal fluorescence microscopy. Figure 2a illustrates the case for 30% Chol. These data reveal a marked 3D ordering of the fluorescence patterns over micrometre depth in sample thicknesses, suggesting near-complete alignment of the domains across hundreds of membrane lamellae. (Supplementary Movie S1 demonstrates the interlamellar ordering of the domains.) Such long-range domain alignment is seen when membrane multilamellae are allowed to equilibrate over time, the length of which depends on the ambient humidity and thickness of the membrane multilamellae; a longer time is required for lower-humidity and thicker samples. The minimum time for complete domain alignment ranges from a few seconds for a 1-µm-thick membrane stack in bulk water to several days for 7-µm-thick samples at 30% relative humidity.

Dynamics of phase separation in stacked lipid bilayers

To gain a mechanistic understanding of interlamellar domain alignment, we began by examining the dynamics in real time (Supplementary Movie S2). These results shed light on several salient aspects of domain alignment. These are discussed in turn below.

First, the dynamics of domain alignment suggests a relationship between the intralayer phase separation and the interlayer alignment. Figure 2b shows confocal images of a membrane multilamellae (relative humidity: 30%) obtained before the complete alignment. Figure 2b(1) illustrates an arbitrarily selected layer (z-slice) revealing circular, well-resolved, L_o phase domains. However, the projection across all layers (Fig. 2b(2)) under these conditions reveals that the domains across different lamellae are not aligned. This indicates that lateral phase separation within single lamellae precedes and begins independently of the domain alignment between the neighbouring lamellae.

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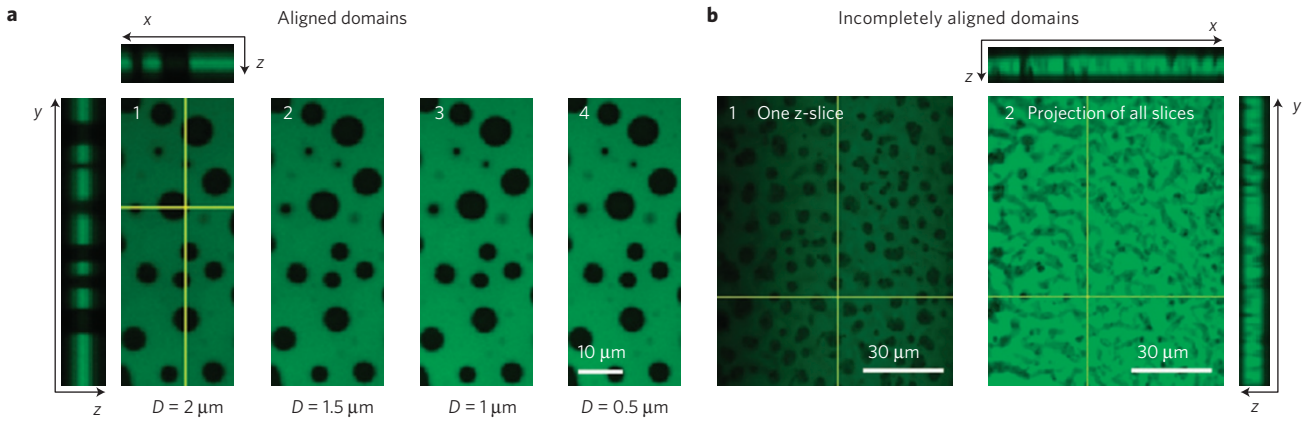

Figure 2 | Depth-resolved confocal fluorescence microscopy of aligned and unaligned stacked lipid bilayers. Membrane multilamellae consisting of (1:1 SM/DOPC + 30%Chol) at 30% relative humidity, doped with Rho-PE (0.1 mol%). **a**, (1–4) aligned domains. The sample thickness is estimated to be 2.5 ± 0.5 μm by confocal microscopy, representing approximately 417 ± 83 layers. Individual *z*-slices at selected depths, *D* (distance from the substrate), reveal near-complete alignment of L_o phase domains over hundreds of layers. The insets (top and left) for (1) show the cross-section view along the yellow-coloured path, indicating near-complete three-dimensional alignment of domains. **b**, (1,2) incompletely aligned domains. Thickness of the sample: 7 ± 1 μm. The image of an arbitrarily selected *z* slice shows well-formed circular domains suggesting in-plane phase separation (1). A projection across all slices shows that the domains in different slices, although formed, are not spatially aligned (2).

Second, comparing average domain sizes, obtained in quasi-equilibrated membrane stacks of widely different thicknesses (under identical preparation conditions), reveals that larger domains form for thicker samples (Fig. 4). This correspondence between the average domain size and the number of lamellae suggests the existence of a coupling between the lamellae driving the domain growth.

Third, it is well known[27] that the temporal evolution of circular domain size in single lipid monolayers or bilayers follows the universal growth law, $\langle A \rangle \sim t^{2n}$, where $\langle A \rangle$ is the average domain area, t is time, and the growth exponent, n, is $<1/3$, consistent with the classical diffusion-limited growth[3,28]. Analysing domain evolution in membrane multilamellae maintained at different humidity conditions, using epifluorescence microscopy, reveals that the domain growth in multilamellae also follows the universal growth law. Comparing the growth rate in several independent experiments, we find that the growth exponent falls in the range of $1/3 < n < 1/2$. A typical example is shown in Fig. 3, for which $n = 0.46$ was obtained. (For fitting, we ignore the initial stages as they characterize the fastest growing unstable mode in the concentration field[29].) These systematically higher values of the growth exponent—compared with the ones seen for diffusion-limited domain formation in single lipid monolayers[3] and bilayers[28]—suggest that the domain growth involves additional contributions beyond diffusional dynamics. We attribute this to originate from the domain–domain coalescence. Such coalescence events are usually rare in domain-forming single lipid monolayers[3] and bilayers[3,28]. In contrast, we find that coalescence occurs rather frequently in membrane multilamellae (Supplementary Movie S3). As we discuss later in the manuscript, this coalescence-driven domain growth results from an interplay between lateral phase separation and interlamellar smectic coupling unique to membrane multilamellae.

Taken together, the three lines of evidence—indicating that: phase-separated seed domains form independently in each layer before the alignment (Fig. 2b); the domain sizes increase with the number of lamellae (Fig. 4a–d); and the domain coalescence plays a key role in domain coarsening (Fig. 3)—establish a remarkable interplay between intralayer phase separation and interlayer coupling driving self-ordered epitaxial films of stacked lipid bilayers.

A kinetic picture, which reconciles these experimental observations, can be drawn, such as described below. Consider that small phase-separated domains form (Fig. 4g(1,2)) in each layer en route

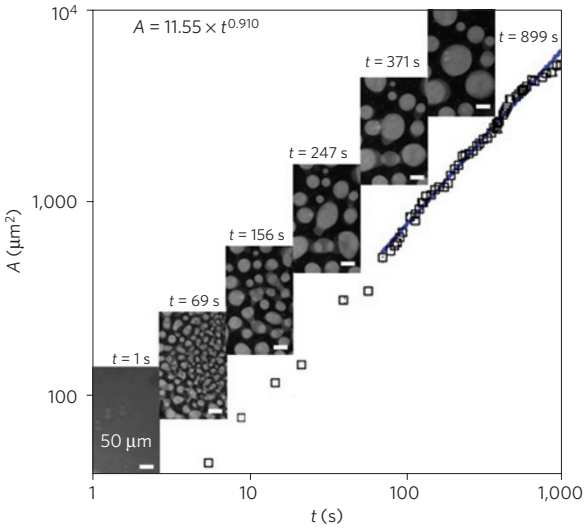

Figure 3 | Domain evolution and coarsening. A representative example of domain-size evolution in stacked lipid bilayers consisting of equimolar SM and $DOPC_1$ at 30% Chol ($Chol_{30\%}/SM_1/DOPC_1$) doped with 0.1 mol% Rho-PE in bulk water. Fitting domain size evolution to the universal growth law, $\langle A \rangle \sim t^{2n}$, yields $n = 0.455$ for this example. Scale bars, 50 μm.

to their equilibration. As these domains grow and coarsen through merging, a higher probability of coalescence is consistent with the formation of larger domains. Next, having a greater number of lamellae increases the overlapping probability of domains across different lamellae. As, according to our experimental result, the domain size increases with the number of lamellae, we can assume that this overlapping probability also increases the coalescence probability through interlamellar bridge formation and hence increases the domain size. Greater numbers of lamellae provide more bridges, which promote the formation of domains with larger surface area (compare Fig. 4g(1,2)). Note that thicker multilayers also introduce longer bridges with more indirect connections. To be effective, such bridges require stronger and long-range interlayer coupling to promote the lateral coalescence of domains. Thus, in this picture, not only the number of lamellae but also the strength of the interlayer coupling influences the growth of the domains through coalescence.

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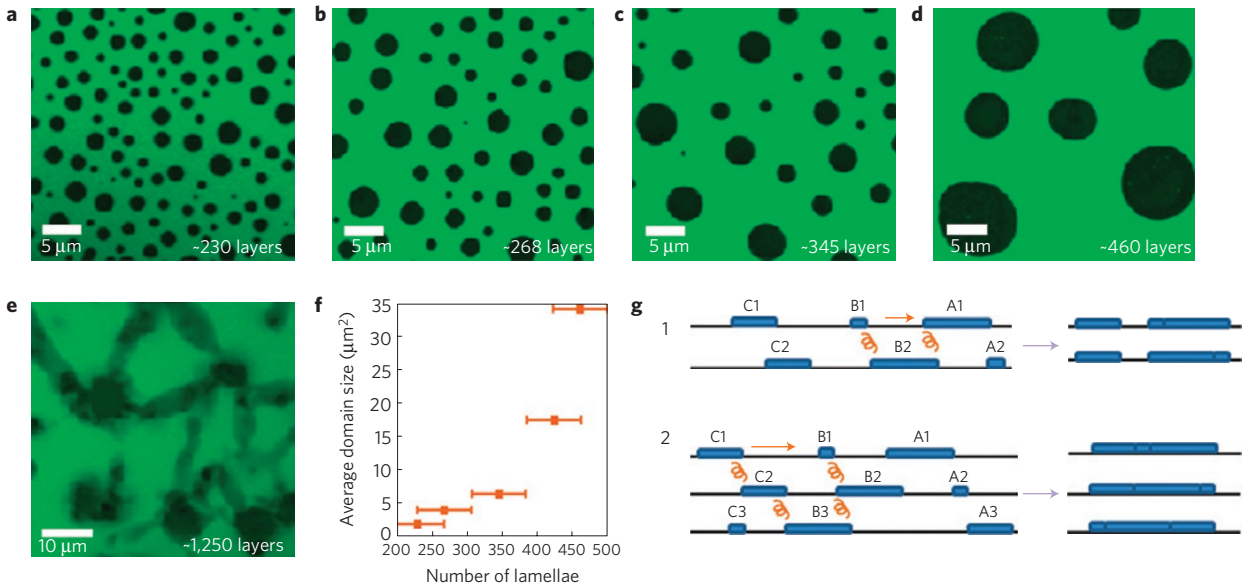

Figure 4 | Dependence of domain size on the number of lamellae. a–e, Projection of *z*-slices in confocal images of multilamellar lipid stacks with different sample thicknesses. All samples followed an identical history and have an identical composition: ($Chol_{30\%}/SM_1/DOPC_1$), 0.1 mol% Rho-PE, 30% relative humidity, 30 days after temperature quenching. The domain size and the time required for complete domain alignment increase with the number of layers. From **a** to **e** the number of lamellae is increasing. The aligned domain area also increases from **a** to **d**; in **e** the alignment has not yet completed at the time of the measurement. **a**, Thickness of multilayer $d = 1,380 \pm 230$ nm, estimated number of lamellae $N = 230 \pm 38$, average domain size $A = 1.90\,\mu m^2$; **b**, $d = 1,610 \pm 230$ nm, $N = 268 \pm 38$, $A = 4.02\,\mu m^2$; **c**, $d = 2,070 \pm 230$ nm, $N = 345 \pm 38$, $A = 6.45\,\mu m^2$; **d**, $d = 2,760 \pm 230$ nm, $N = 460 \pm 38$, $A = 34.21\,\mu m^2$; **e**, $d = 7,500 \pm 230$ nm, $N = 1,250 \pm 38$. **f**, Dependence of average domain size on the number of lamellae in the membrane stack. **g**, (1,2) a schematic depicting a kinetic model for the dependence of domain size on the number of lamellae. The model proposes a cooperative epitaxy through a bridge mechanism by which interlamellar coupling induces domain growth and coarsening. (1) depicts a nearest-neighbour interaction in a two-layer structure. Here, one bridge is shown: the uncorrelated domains in a given layer, A1 and B1, are drawn into a larger domain by the interlamellar interactions with the domain B2 in the second layer. As there is no bridge between B1 and C1, C1 does not join A1 and B1. However, in a three-layer structure, such as (2), interaction of the second and third layers makes a new bridge (C1–C2–B3–B2–B1), which drags C1 to B1 (and A1) and ultimately forms larger domains. (Only one bridge in each panel is shown.)

Topographical mismatch in stacked bilayers

The interlamellar alignment of L_o and L_d domains reported above raises an intriguing question about the molecular organization of the lipids in the phase-separated membrane multilayers: how does the hydrophobic mismatch between the ordered and taller L_o domains and shorter, disordered L_d phase manifests in such phase-separated morphologies? It is known that the Chol- and SM-enriched L_o domains have a 0.6–0.8 nm higher hydrophobic thickness than the surrounding L_d phase in single lipid bilayers[30,31]. In the multilamellar samples investigated here ($N = 100$–$1,200$ layers), this height disparity necessarily scales with the number of layers, either amplifying the overall topographic mismatch or stabilizing the binary patterns of interlamellar hydration (Fig. 5b).

To address this question, we first carried out atomic force microscopy (AFM) measurements. A comparison of the AFM images of phase-separated membrane multilamellae consisting of a varying number of lamellae reveals two noteworthy features: there is a noticeable height difference between L_o and L_d domains, and the height difference between the two domains increases with the number of lamellae (Fig. 5a(1,2)). These observations lead us to propose a model for the interlamellar organization of domains, discussed below. Figure 5b(1,2) shows two possible organizations that the domains may adopt. In the first configuration (Fig. 5b(1)), the thickness of the interbilayer water in the two L_o and L_d regions is allowed to adjust so that the two phases are at the same topographic level. In the second configuration (Fig. 5b(2)), the thickness of the interbilayer water is maintained to be comparable for the two regions. In the latter model, L_o domains should appear taller than the L_d domains in the lipid multilayers with the height difference, which increases with the number of lamellae. The observed correlation between the height difference of the domains and the number of lamellae in our AFM measurements is consistent with the latter, second configuration (Fig. 5b(2)), further supported by X-ray diffraction experiments described below.

To further characterize the smectic alignment of phase-separated domains, we carried out X-ray diffraction experiments using comparable multilamellar membranes. Figure 6 shows an example of X-ray diffraction data obtained using a CuKα-tube-source in-house X-ray reflectometer for a multilayer membrane consisting of a 1:1 DPPC(d62)/DOPC mixture doped with 16% Chol at 98% relative humidity. For these measurements, we substituted SM in our raft-forming mixtures with higher-purity DPPC. Previous studies using giant vesicles establish that L_o–L_d phase separation comparable to the SM-based ternary mixtures also occurs in DPPC mixtures containing DOPC and Chol[19–22]. As expected, we find that above the phase-separation temperature (37 °C), membrane multilamellae produce a homogeneous phase with a single repeat unit spacing (*d*-spacing) of 55.4 Å (estimated from the third-order diffraction peak), which represents the average thickness of the homogeneous lipid bilayer plus the water layer thickness. When the temperature is lowered below the phase-transition temperature (24 °C), two separate phases emerge as indicated by the appearance of two separate sets of diffraction peaks. From the position of these diffraction peaks, the *d*-spacing for one phase (L_o) is 62.7 Å and for the second phase (L_d) is 54.6 Å, consistent with coexistent L_o and L_d phases[32].

The appearance of two sets of diffraction peaks with two well-defined *d*-spacings confirms the formation of two separate phases aligned within the multilamellar sample. The coherence length of each aligned domain can be calculated from the width of the corresponding peaks (after instrumental resolution correction). For

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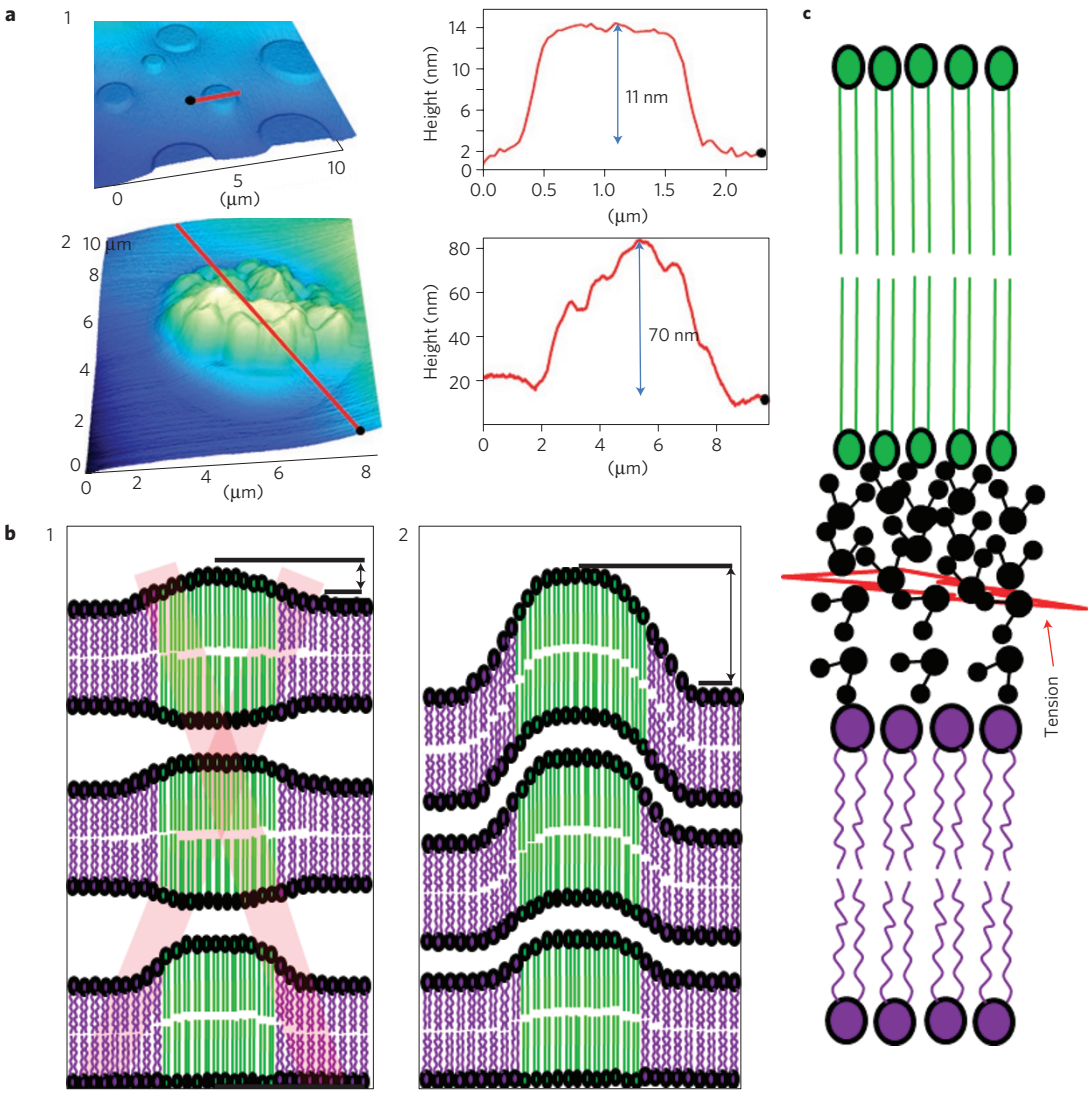

Figure 5 | Proposed molecular explanation of interlamellar coupling of domains in stacked lipid bilayers. a, AFM images of two arbitrarily selected multilayers characterized by a difference in their average thicknesses (1,2). $d(2) > d(1)$. A comparison of the height differences between the domains and the surrounding phase between (1),(2) reveals a correlation with the thickness (or number of lamellae) of the multilayers under identical sample conditions. Both samples (1),(2) have an identical history and composition: ($Chol_{30\%}/SM_1/DOPC_1$) doped with 0.1 mol% Rho-PE at 30% relative humidity. The thickness of multilayer (1) is $1,380 \pm 230$ nm and that of (2) is $2,760 \pm 230$ nm. The average domain height difference across several samples with thicknesses comparable to that of samples (1),(2) lies in the range of 5–25 nm and 45–100 nm, respectively. **b**, A schematic representation (two types of lipid domains are shown in purple and green; arrows denote the height difference between them) of two plausible configurations (1,2) for the organization of layers. On the basis of these data (1,2), the organization of scheme (2) seems to be preferred (see the text for details). **c**, A schematic depiction of how interlamellar mismatch of domains can give rise to a mismatch in the water network adjacent to the domains, which in turn creates epistructural tension of the water.

all of the samples that we measured, the number of vertically aligned layers reached the maximum measurable, with our instrumental resolution, in the range of 100–400 bilayers, confirming long-range columnar ordering of L_o (as well as L_d) domains. Repeating the experiment with a pure DOPC multilayer showed that the number of coherently scattering lamellae is comparable to that of the phase-separated multilayers above. Note that these results do not provide information regarding the lateral organization of individual molecules within the registered bilayer units, which is not expected because of long-range lateral fluidity in individual bilayers. Moreover, it is important to note also that the actual number of aligned layers may be greater than our estimate of 100–400 layers because the aligned domains could lose coherence after a certain number of layers owing to localized defects, which manifests in the finite width of the diffraction peaks.

Plausible role of epistructural tension of water

The immiscibility of the L_d and L_o phases in monolayers and bilayers reflects the existence of line tension arising from the hydrophobic mismatch between the molecules in the L_d and L_o domains[33,34]. A comparison of the two plausible configurations (Fig. 5b) reveals that domain alignment does not reduce hydrophobic mismatch of tails. On the basis of these considerations, we speculate that, in membrane multilamellae, the contribution of the aqueous interfaces of the hydrophilic headgroups to the effective line tension is more pronounced and the alignment in the z direction must be controlled by the properties of the water–headgroup interface in apposing bilayer units.

It is well known that the interaction of water molecules with the lipid headgroups markedly disrupts the normal hydrogen-bonding network associated with the bulk water[35]. This in turn

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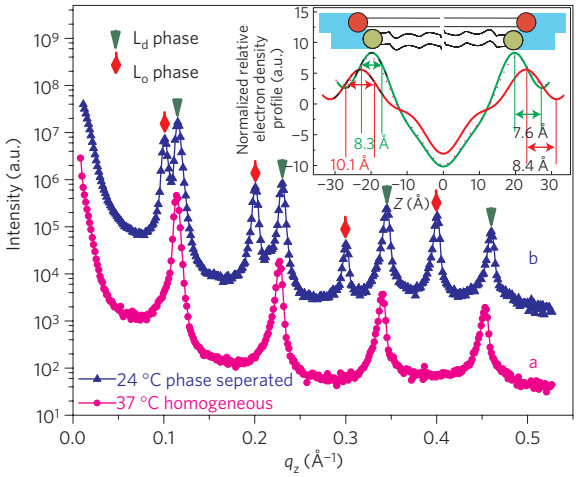

Figure 6 | X-ray diffraction analysis. X-ray scattering profiles for a raft-forming ternary stacked lipid bilayer consisting of 1:1 DPPC(d62)/DOPC + 16% Chol at 98% relative humidity. Above the phase-transition temperature, the membrane multilamellae is in a homogeneous phase (curve a, pink) characterized by a single set of diffraction peaks. By decreasing the temperature to below the phase transition, a second set of diffraction peaks appears (curve b, blue), which indicates the appearance of two separate periodicities consistent with the interlayer alignment of coexisting intralayer L_o and L_d phases. The Gaussian fits of the diffraction peaks provide an estimate for the number of layers aligned in the *z* direction. In this specific sample, the number of coherently scattering layers in L_o and L_d phases are about 200 and 300 layers, respectively. The inset shows the normalized relative electron density profile of the unit cell constructed from the diffraction data, with a schematic illustration of the L_o and L_d phases on top. The pink dotted line is the electron density of the homogeneous phase at 37 °C; the red line and the green line are the electron density profiles for the L_o and L_d phases, respectively, at 24 °C. The distances between the centre of the headgroup and the centre of the water layer are measured to be 8.4 Å and 7.6 Å for the L_o and L_d phases, respectively (right), which gives the distances between the centres of adjacent headgroups separated by the water layers to be 16.8 and 15.2 Å. The Gaussian fits to the headgroup sizes give full-width at half-maximum of 10.1 and 8.3 Å for the L_o and L_d phases, respectively (left).

induces strong ordering and an orientational change of water molecules within about 1 nm from the headgroups, modifying the structure of the hydrogen-bonding network[35–37]. In fact, it is now well established that the interbilayer hydration force in a multilayer membrane is the result of such ordering and the partial orientation of interlamellar water[38–42]. Furthermore, it is also known that the hydration properties of the L_o and the L_d phases are significantly different from each other[43]. As a result, water binding characteristics at the aqueous interfaces of the L_d and the L_o phases differ considerably[43]. This should then introduce differences in the organization of water molecules and orientation at L_o and L_d phases, as depicted schematically in Fig. 5c. Taken together, the three interfacial features, namely the altered and highly ordered water hydrogen-bonding network, which can extend to about 1 nm from the headgroups of each layer; the 1–2 nm interlamellar distance between headgroups of apposing layers depending on the ambient humidity; and differences in the water networks for the L_o and L_d phases, suggests that L_d and L_o domains produce distinctly different lipid–water interfaces for the domains and the surroundings. The mismatch of the water network necessarily introduces an additional epistructural surface-tension[37,44,45] penalty at the interface where the two networks join, which is energetically unfavorable. Domain alignment reduces such interfaces (to only in-plane interfaces of L_o and L_d phases) and thus lowers the overall free energy of the system, suggesting a plausible mechanism for the interlamellar domain ordering observed here (Fig. 5c).

The intensities of the X-ray diffraction peaks from each of the two coexisting phases were used to reconstruct the electron density profile of the bilayer in the normal direction for each phase, following previously established methods and employing similar criteria for choosing the phases of the Bragg reflections[46,47] (see Supplementary Information S4 for details). These are presented relative to the average electron density in the inset to Fig. 6 with an arbitrary scale factor for each. Two key conclusions may be drawn from these profiles. First, the distances between the centres of adjacent headgroups separated by the water layers are 16.8 Å for the L_o phase (red) and 15.2 Å for the L_d phase (green). Thus, the average thicknesses of the water layers in the two phases differ by only 1.6 Å, which is almost negligible compared to the total water layer. This in turn lends further support to our foregoing inference (based on AFM data above) for the preference for the scenario depicted in Fig. 5b(2). Second, our Gaussian fits to the headgroup profiles show that the full-width at half-maxima of those in the L_o phase (red) and those in the L_d phase (green) are 10.1 Å and 8.3 Å respectively. As the phosphocholine headgroups for the lipids in the two coexisting phases are identical, we interpret this difference in the sizes of the headgroup region in terms of a roughness difference. This subtle, but reproducible, roughness difference—together with the 1.6 Å difference of water layer thickness—may reflect the difference in the water hydrogen-bonding network between the L_o and the L_d phases. This is because the water molecules in the confined region between lipid bilayers are highly ordered and in direct contact with the hydrophilic headgroups. A detailed analysis of the X-ray diffraction data and of the characterization of the nature of the boundary between the coexisting phases is well outside the scope of the present paper and will be reported separately (Y. M. *et al.*, manuscript in preparation, 2012).

The observed spontaneous ordering of laterally phase-separated domains in stacked lipid bilayers reported here offers several important features that might prove useful for practical applications of membrane-based functional materials. First, membrane multilamellae are structurally robust, exhibiting long-term stability under non-oxidizing ambient conditions, and are amenable to further chemistry such as at the lipid–water interface for further improvements in structural integrity. Second, they allow programming of domain sizes within which membrane-associated functions can be stacked: controllable sizes of domains—ranging from a few nanometres to several hundred micrometres—can be produced by changing the number of lamellae and/or the ambient humidity. Third, the three-dimensionally aligned phase-separated domains can form and maintain their structural integrity under variable, including low, humidity levels.

Methods

Preparation of membrane multilamellae. Membrane multilamellae at solid surfaces are prepared by first depositing 1–3 μl droplets of stock solutions of the desired lipid mixture (lipid concentration, ~5–20 mg ml^{-1}; solvent, chloroform or ethanol/water 4:1 volume ratio) on freshly cleaned glass slides and allowing the solvent to evaporate under vacuum overnight. To investigate domain formation in bulk water, dry plaques of lipids so obtained on glass slides are incubated under 100% relative humidity at a temperature between 50 °C and 60°C for at least two days by adapting previously reported methods[17,18]. (As this temperature range is above the miscibility transition temperatures for the lipid mixtures investigated, domains do not form during incubation.) A slow and gradual hydration during this incubation period results in well-oriented, smectic membrane multilamellae[17,18]. Subsequently, the sample is immersed in bulk deionized water at room temperature. For experiments performed at partial humidities, the samples are incubated at desired ambient humidities and temperature 50–60 °C using controlled-humidity sample chambers. The domain structure and dynamics at different relative humidities are examined by quenching the sample temperatures to room temperature, ~22°C.

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Characterization. Samples are characterized using a combination of epifluorescence microscopy, atomic force microscopy, confocal fluorescence microscopy measurements and X-ray diffraction. Epifluorescence measurements were performed using an inverted epifluorescence microscope (Nikon TE200E, Technical Instruments), equipped with a high-spatial-resolution CCD (charge-coupled device) camera. Confocal and atomic force microscopy were performed using a combined system in which an AFM scanner (Asylum Research MFP-3D) is mounted to the stage of the laser scanning confocal (Olympus (IX-81) America FluoView 1000). For AFM, AC mode imaging using an Olympus AC 240 lever with a spring constant $k \approx 2$ allowed for the height (topography) imaging. Image analyses were performed using NIH ImageJ. X-ray reflectivity measurements were performed using a Cu Kα-tube-source four-circle X-ray reflectometer with a Bicron scintillation X-ray detector. The sample was mounted in a constant-humidity cell containing a reservoir of saturated K_2SO_4 solution yielding a relative humidity of 98%. Details of electron density analyses are provided in the Supplementary Information.

Received 26 February 2011; accepted 11 September 2012; published online 21 October 2012

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Acknowledgements

This work is supported by a grant from the Biomolecular Materials Program, Division of Materials Science and Engineering, Basic Energy Sciences, US Department of Energy under Award no. DE-FG02-04ER46173.

Author contributions

L.T., Y. M. and G.C. performed experiments, analysed data and wrote initial drafts of the manuscript. D.V. contributed extensively to all experimental data analyses and helped L.T. with the theoretical model. A.N.P. and S.K.S. conceived the overall project, provided intellectual and technical guidance, performed final edits of the manuscript and are principal investigators of the supporting grant. All authors contributed to writing and revising the manuscript, and agreed on its final content.

Additional information

Supplementary information is available in the online version of the paper. Reprints and permissions information is available online at www.nature.com/reprints. Correspondence and requests for materials should be addressed to S.K.S. or A.N.P.

Competing financial interests

The authors declare no competing financial interests.

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